FAULT CODE 324 (QSK38) - Injector Solenoid Driver Cylinder 3 Circuit - Current Below Normal or Open Circuit

Warnings and Cautions

⚠ WARNING ⚠

The injector solenoids receive high voltage when the engine is operating. To reduce the possibility of personal injury from electrical shock, do not wear jewelry or damp clothing, and do not touch the injector solenoids or the solenoid wires when the engine is operating.

⚠ CAUTION ⚠

To reduce the possibility of damaging a new engine control module (ECM), all other active fault codes must be investigated prior to replacing the ECM.

⚠ CAUTION ⚠

To reduce the possibility of pin and harness damage, use the following test leads when taking a measurement: Part Number 3824811 - male Deutsch™ test lead, Part Number 3824812 - female Deutsch™ test lead, Part Number 3822758 - male Deutsch™/AMP™/Metri-Pack™ test lead, and Part Number 3822917 - female Deutsch™/AMP™/Metri-Pack™ test lead.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check for active fault codes.	
	STEP 1A. Read the fault codes with INSITE™ electronic service tool.	Fault Code 322, 323, or 324 active?
	STEP 1B. Read the fault codes with INSITE™ electronic service tool.	Fault Code 324 not active?

STEPS		SPECIFICATIONS
	STEP 1C. Read the fault codes with INSITE™ electronic service tool.	Multiple injector fault codes active?
STEP 2.	Check the injector and injector solenoid driver cylinder 3 circuit for an open circuit.	
	STEP 2A. Inspect the engine harness connections.	Connectors properly connected?
	STEP 2A-1. Inspect the engine harness and ECM connector pins.	Dirty or damaged pins?
	STEP 2B. Check for an open circuit.	0.5 to 5 ohms?
	STEP 2C. Inspect the injector harness connector pins.	Dirty or damaged pins?
	STEP 2D. Check for an open circuit.	0.5 to 5 ohms?
	STEP 2E. Read the fault codes.	Fault Code 324 is the only active fault code?
STEP 3.	Check the engine harness.	
	STEP 3A. Inspect the engine harness and injector solenoid driver connector pins.	Dirty or damaged pins?
	STEP 3B. Check the injector solenoid drivers for short circuits to ground.	Greater than 100k ohms?
	STEP 3C. Inspect the engine harness.	Dirty or damaged pins, or damaged wire insulation?
	STEP 3C-1. Check the engine harness for short circuit to ground.	Greater than 100k ohms?
	STEP 3C-2. Check the engine harness for pin-to-pin short circuits.	Greater than 100k ohms?
STEP 4.	Clear the fault codes.	
	STEP 4A. Disable the fault code.	Same multiple injector fault codes active?
	STEP 4B. Clear the inactive fault codes.	All fault codes cleared?

STEP 1. Check for active fault codes.

STEP 1A. Read the fault codes with INSITE™ electronic service tool.

Conditions :

- Turn keyswitch ON.
- INSITE™ electronic service tool connected.

Action	Specification/Repair	Next Step
Operate the engine and observe the fault codes. • Use INSITE™ electronic service tool to clear the fault codes. • Start the engine and let it idle for 1 minute. • Use INSITE™ electronic service tool to read the fault codes.	Fault Code 322, 323, or 324 active? **YES**	**1B**
	Fault Code 322, 323, or 324 active? **NO**	**Use the following procedure for inactive or intermittent fault code. Refer to Procedure 019-362 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-362.html)**

STEP 1B. Read the fault codes with INSITE™ electronic service tool.

Conditions :

- Turn keyswitch ON.
- INSITE™ electronic service tool connected.

Action	Specification/Repair	Next Step
Read the fault codes. • Use INSITE™ electronic service tool to read the fault codes.	Fault Code 324 **only** active? **YES**	**2A**
	Fault Code 324 **only** active? **NO**	**1C**

STEP 1C. Read the fault codes with INSITE™ electronic service tool.

Conditions :

- Turn keyswitch ON
- INSITE™ electronic service tool connected.

Action	Specification/Repair	Next Step

Read the fault codes. • Use INSITE™ electronic service tool to read the fault codes.	Multiple injector fault codes active? **YES**	**3A**
	Multiple injector fault codes active? **NO**	**2A**

STEP 2. Check the injector and injector solenoid driver cylinder 3 circuit for an open circuit.

STEP 2A. Inspect the engine harness connections.

Conditions :
- Turn keyswitch OFF.

Action	Specification/Repair	Next Step
Make sure the following engine harness connections are properly made: • Connect engine harness to the ECM • Connect engine harness to the injector.	Connectors properly connected? **YES**	**2A-1**
	Connectors properly connected? **NO** **Repair :** Install the engine harness connectors properly.	**4A**

STEP 2A-1. Inspect the engine harness and ECM connector pins.

Conditions :
- Turn keyswitch OFF
- Disconnect engine harness connector from the ECM 60 pin connector.

Action	Specification/Repair	Next Step

<table>
<tr>
<td rowspan="2">Inspect the engine harness and ECM connector pins for the following:
Loose connector
Corroded pins
Bent or broken pins
Pushed back or expanded pins
Moisture in or on the connector
Missing or damaged connector seals
Dirt or debris in or on the connector pins
Connector shell broken
Wire insulation damage
Damaged connector locking tab.
Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)</td>
<td>Dirty or damaged pins?
YES

Repair :
Refer to the circuit diagram or wiring diagram for all harness interconnections.
A damaged connection has been detected in the ECM connector or engine harness.
Repair the damaged pins. Repair or replace the engine harness or replace the ECM, whichever has the damaged pins.
Flush the dirt, debris, or moisture from the connector pins, use electronic contact cleaner, Part Number 3824510.
Install the appropriate connector seal if it is damaged or missing.
Repair the engine harness. Refer to Procedure 019-204 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-204.html)
Replace the ECM. Refer to Procedure 019-031 in Section 19. (/qs3/pubsys2/xml/en/procedures/105/105-019-031.html)</td>
<td>4A</td>
</tr>
<tr>
<td>Dirty or damaged pins?
NO</td>
<td>2B</td>
</tr>
</table>

STEP 2B. Check for an open circuit.

Conditions :

- Turn keyswitch OFF
- Disconnect engine harness connector from the ECM 60 pin connector.

Action	**Specification/Repair**	**Next Step**
Check for continuity in the injector solenoid driver cylinder 3 circuit. • Measure the resistance between the injector solenoid driver cylinder 3 SIGNAL pin and the injector solenoid driver cylinder 3 RETURN pin at the engine harness ECM connector. Use a wiring diagram for connector pin identification and use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	0.5 to 5 ohms? **YES**	**2E**
	0.5 to 5 ohms? **NO**	**2C**

STEP 2C. Inspect the injector harness connector pins.

Conditions :

- Turn keyswitch OFF
- Disconnect engine harness connector from the ECM 60 pin connector
- Disconnect engine harness connector from the injector solenoid driver cylinder 3 connector.

Action	**Specification/Repair**	**Next Step**

Inspect the engine harness and injector connector pins for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? **YES** **Repair :** Clean the connector and pins. Replace the damaged section of the harness or damaged injector. Refer to the circuit diagram or wiring diagram for all harness interconnections. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html) For QSK45 and QSK60 engines, see the following procedure to replace the injector in Service Manual, QSK45 and QSK60, Bulletin 4021530. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html) For K38, K50, and QSKK38 engines, see the following procedure to replace the injector in Service Manual, K38, K50, QSK38 and QSK50, Bulletin 4021528. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html)	**4A**
	Dirty or damaged pins? **NO**	**2D**

STEP 2D. Check for an open circuit.

Conditions :		
- Turn keyswitch OFF - Disconnect engine harness connector from the ECM 60 pin connector - Disconnect engine harness connector from the injector solenoid driver cylinder 3 connector.		
Action	**Specification/Repair**	**Next Step**

Check for continuity in the injector. • Measure the resistance between the injector solenoid driver cylinder 3 SIGNAL pin and the injector solenoid driver cylinder 3 RETURN pin at the injector connector. Use a wiring diagram for connector pin identification and use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	0.5 to 5 ohms? **YES** **Repair :** Troubleshoot all harnesses connected in series to determine which contains the open circuit. Refer to the circuit diagram or wiring diagram for all harness interconnections. Replace the damaged section of the harness. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html)	**4A**

	0.5 to 5 ohms? **NO** **Repair :** Replace the injector(s). For QSK45 and QSK60 engines, see the following procedure to replace the injector in Service Manual, QSK45 and QSK60, Bulletin 4021530. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html) For K38, K50, and QSKK38 engines, see the following procedure to replace the injector in Service Manual, K38, K50, QSK38 and QSK50, Bulletin 4021528. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html)	**4A**

STEP 2E. Read the fault codes.

Conditions :

- Connect all components.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step

Operate the engine and observe the fault codes. • Use INSITE™ electronic service tool to clear the fault codes. • Start the engine and let it idle for 1 minute. • Use INSITE™ electronic service tool to read the fault codes.	Fault Code 324 is the **only** active fault code? **YES** **Repair :** • Replace the ECM. Refer to Procedure 019-031 in Section 19. (/qs3/pubsys2/xml/en/procedures/105/105-019-031.html)	**4A**
	Fault Code 324 is the **only** active fault code? **NO**	**4A**

STEP 3. Check the engine harness.

STEP 3A. Inspect the engine harness and injector solenoid driver connector pins.

Conditions :

- Turn keyswitch OFF.
- Disconnect engine harness connectors from the injector solenoid driver cylinder 1, 3 and 5 connectors.

Action	Specification/Repair	Next Step

Inspect the engine harness and injector connector pins for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab . Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? **YES** **Repair :** Clean the connector and pins. Replace the damaged section of the harness or damaged injector. Refer to the circuit diagram or wiring diagram for all harness interconnections. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html) For QSK45 and QSK60 engines, see the following procedure to replace the injector in Service Manual, QSK45 and QSK60, Bulletin 4021530. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html) For K38, K50, and QSKK38 engines, see the following procedure to replace the injector in Service Manual, K38, K50, QSK38 and QSK50, Bulletin 4021528. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html)	**4A**
	Dirty or damaged pins? **NO**	**3B**

STEP 3B. Check the injector solenoid drivers for short circuits to ground.

Conditions :

- Turn keyswitch OFF.
- Disconnect engine harness connectors from injector solenoid driver cylinder 1, 3, and 5 connectors.

Action	Specification/Repair	Next Step
Check for a short circuit to ground. • Measure the resistance between the injector solenoid driver cylinder 1 SIGNAL pin and engine block ground. • Measure the resistance between the injector solenoid driver cylinder 3 SIGNAL pin and engine block ground. • Measure the resistance between the injector solenoid driver cylinder 5 SIGNAL pin and engine block ground. Use a wiring diagram for connector pin identification and use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Greater than 100k ohms? **YES**	**3C**
	Greater than 100k ohms? **NO** **Repair :** Replace the injector(s). For QSK45 and QSK60 engines, see the following procedure to replace the injector in Service Manual, QSK45 and QSK60, Bulletin 4021530. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html) For K38, K50, and QSKK38 engines, see the following procedure to replace the injector in Service Manual, K38, K50, QSK38 and QSK50, Bulletin 4021528. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html)	**4A**

STEP 3C. Inspect the engine harness.

Conditions :

- Turn keyswitch OFF
- Disconnect engine harness connectors from the injector solenoid driver cylinder 1, 3, and 5 connectors
- Disconnect engine harness connectors from the ECM 60 pin connector.

Action	Specification/Repair	Next Step

<table>
<tr>
<td>Inspect the engine harness and ECM connectors for the following:
Loose connector
Corroded pins
Bent or broken pins
Pushed back or expanded pins
Moisture in or on the connector
Missing or damaged connector seals
Dirt or debris in or on the connector pins
Connector shell broken
Wire insulation damage
Damaged connector locking tab.
Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)</td>
<td>Dirty or damaged pins or damaged wire insulation?
YES

Repair :
Clean the connector and pins.
Replace the damaged section of the harness or damaged injector.
Refer to the circuit diagram or wiring diagram for all harness interconnections.
Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html)
For QSK45 and QSK60 engines, see the following procedure to replace the injector in Service Manual, QSK45 and QSK60, Bulletin 4021530. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html)
For K38, K50, and QSKK38 engines, see the following procedure to replace the injector in Service Manual, K38, K50, QSK38 and QSK50, Bulletin 4021528. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html)</td>
<td>4A</td>
</tr>
</table>

	Dirty or damaged pins or damaged wire insulation? **NO**	**3C-1**

STEP 3C-1. Check the engine harness for short circuit to ground.

Conditions :

- Turn keyswitch OFF.
- Disconnect engine harness connectors from the ECM 60 pin connector.
- Disconnect engine harness connectors from the injector solenoid driver cylinder 1, 3, and 5 connectors.

Action	Specification/Repair	Next Step

Check for a short circuit to ground. • Measure the resistance from the injector solenoid driver cylinder 1 SIGNAL pin at the engine harness ECM 60 pin connector to engine block ground. Repeat the check at the injector solenoid driver cylinder 3 SIGNAL and the injector solenoid driver cylinder 5 SIGNAL pins at the engine harness ECM 60 pin connector. • Measure the resistance from the injector solenoid driver cylinder 1 RETURN pin at the engine harness ECM 60 pin connector to engine block ground. Repeat the check for the injector solenoid driver cylinder 3 RETURN and the injector solenoid driver cylinder 5 RETURN pins at the engine harness ECM 60 pin connector. Use a wiring diagram for connector pin identification and use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Greater than 100k ohms? **YES**	**3C-2**
	Greater than 100k ohms? **NO** **Repair :** Troubleshoot all harnesses connected in series to determine which contains the open circuit. Refer to the circuit diagram or wiring diagram for all harness interconnections. Replace the damaged section of the harness. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html)	**4A**

STEP 3C-2. Check the engine harness for pin-to-pin short circuits.

Conditions :

- Turn keyswitch OFF.
- Disconnect engine harness connectors from the ECM 60 pin connector.
- Disconnect engine harness connectors from the injector solenoid driver cylinder 1, 3, and 5 connectors.

Action	Specification/Repair	Next Step

Check for a short circuit from pin-to-pin. - Measure the resistance from the injector solenoid driver cylinder 1 SIGNAL pin at the engine harness ECM 60 pin connector to all other pins in the connector. Repeat the check at the injector solenoid driver cylinder 3 SIGNAL and the injector solenoid driver cylinder 5 SIGNAL pins at the engine harness ECM 60 pin connector. - Measure the resistance from the injector solenoid driver cylinder 1 RETURN pin at the engine harness ECM 60 pin connector to all other pins in the connector. Repeat the check for the injector solenoid driver cylinder 3 RETURN and the injector solenoid driver cylinder 5 RETURN pins at the engine harness ECM 60 pin connector. Use a wiring diagram for connector pin identification and use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Greater than 100k ohms? **YES**	**4A**
	Greater than 100k ohms? **NO** **Repair :** Troubleshoot all harnesses connected in series to determine which contains the pin-to-pin short. Refer to the circuit diagram or wiring diagram for all harness interconnections. Replace the damaged section of the harness. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html)	**4A**

STEP 4. Clear the fault codes.

STEP 4A. Disable the fault code.

Conditions :

- Connect all components.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Disable the fault code. • Start the engine and idle engine for 1 minute. • Use INSITE™ electronic service tool to verify that the fault codes are inactive.	Same multiple injector fault codes active? **YES** **Repair :** Verify that all steps have been completed. If all steps have been completed, then follow the technical escalation process.	**Escalate or call for assistance.**
	Same multiple injector fault codes active? **NO**	**4B**

STEP 4B. Clear the inactive fault codes.

Conditions :

- Connect all components.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step

Clear the inactive fault codes. • Use INSITE™ electronic service tool to clear the inactive fault codes.	All fault codes cleared? **YES**	**Repair complete**
	All fault codes cleared? **NO** **Repair :** Troubleshoot any remaining fault codes.	**Appropriate troubleshooting steps**

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